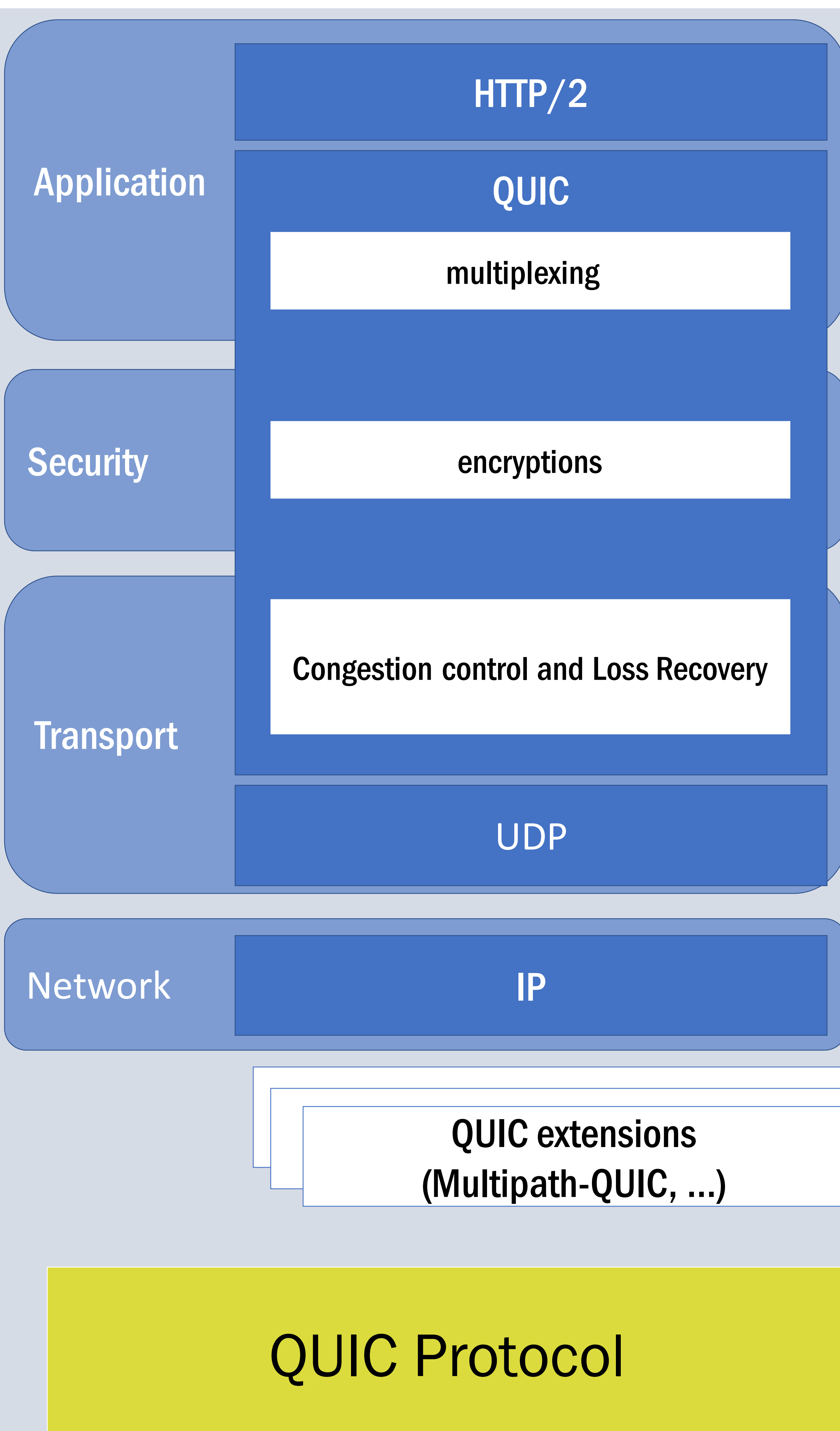


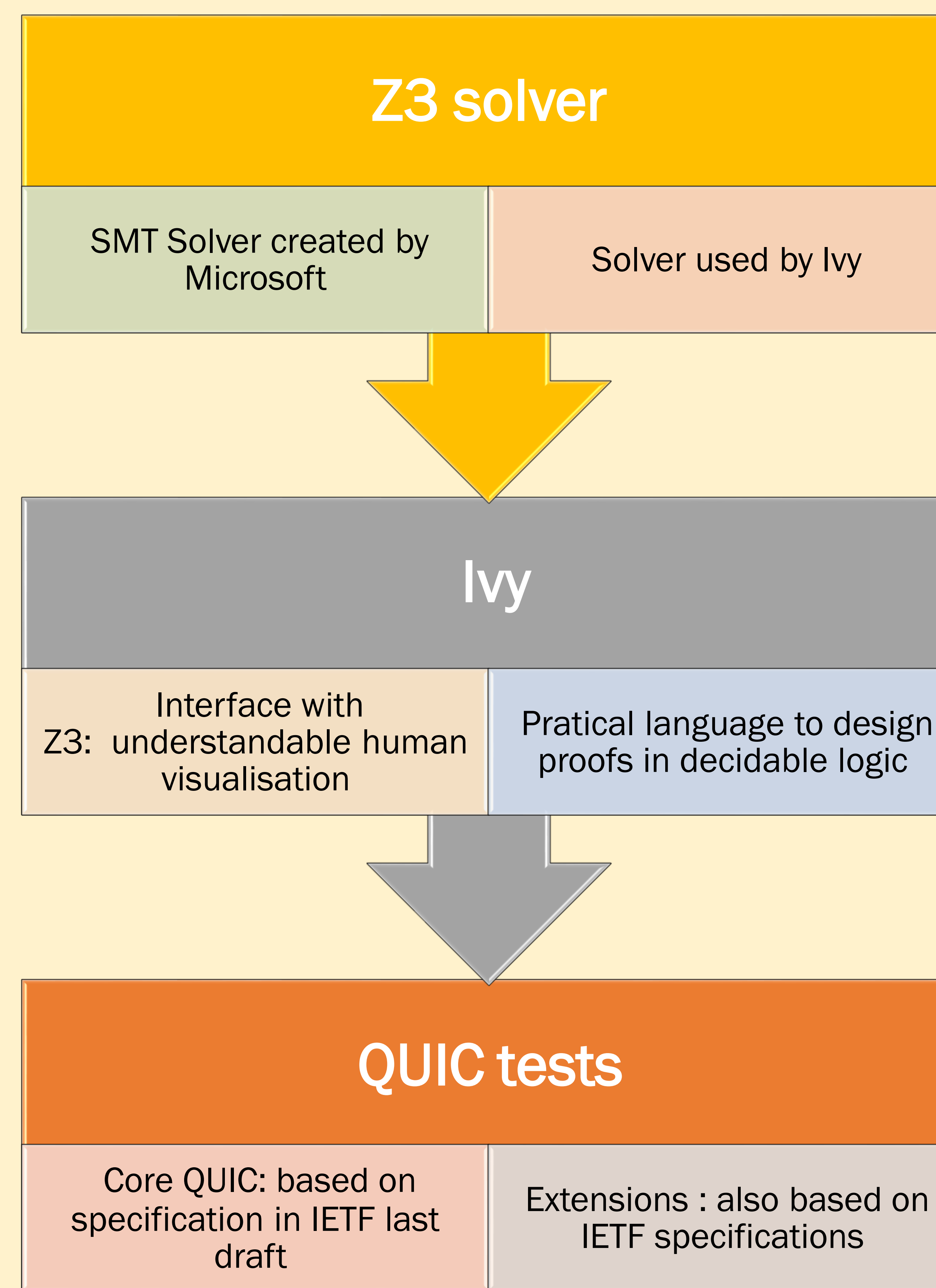
Use of an automated tool for a formal verification of the QUIC protocol and extensions : Ivy

References

- [1] Formal specification and testing of QUIC KL McMillan, LD Zuck - SIGCOMM 2019, <http://mcmil.net/pubs/SIGCOMM19.pdf>
- [2] <https://datatracker.ietf.org/wg/quic/documents/>
- [3] <https://www.multipath-quic.org>
- [4] <https://www.pquic.org>



Automated solver language : Ivy



What we have done

- Reproduce the previous results of McMillan paper based on QUIC draft 18
- Test compatibility of various QUIC implementations with Ivy tool
- Understanding how to use Ivy

What we are doing

- Refactor and adapt to latest QUIC draft
- Check the possibility of verification of properties that has never been verified

What we still have to do

- Make test for QUIC extensions
- Test the corectness of our proofs by the use of a QUIC packet manipulator : QUIC tracker